

Modern Screening Designs

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Paradigm shift

- Traditional experimentation
 - Phase 1: screening experiment where every factor is studied at 2 levels
 - Phase 2: response surface experiment where every factor is studied at 3 levels (or more)
- Tendency to combine screening and response surface experimentation in a single step
- Focus on main effects + interaction effects + quadratic effects
- Designs:
 - Definitive screening designs
 - OMARS designs

Classical screening designs

- Every factor is studied at two levels
- Full factorial designs involve 2^k runs or tests when k factors are studied (= optimal designs)
- Fractional factorial designs need to be used when k is large (= optimal designs for main effects)
 - Small designs: main effects are aliased with two-factor interactions
 - Larger designs: two-factor interactions are aliased with each other
- Plackett-Burman (= optimal designs for main effects)
- Center runs to assess whether there are quadratic effects

Optimal screening designs

- The number of tests you can afford is not always a multiple of 4 or a power of 2
- In that case, you need to compute an optimal design for your number of tests
- Number of tests usually too small to estimate all the two-factor interaction effects
- Two options:
 - Plain optimal design for main effects model
 - Advanced optimal design focusing on main effects but not ignoring interaction effects (Bayesian optimal design, alias optimal design, model-robust design, ...)

Alias optimal design

- Assumed correct model

$$\mathbf{Y} = \mathbf{X}_1 \boldsymbol{\beta}_1 + \mathbf{X}_2 \boldsymbol{\beta}_2 + \varepsilon$$

- Estimated model

$$\mathbf{Y} = \mathbf{X}_1 \boldsymbol{\beta}_1 + \varepsilon$$

- Expectation least squares estimator

$$\begin{aligned} E(\hat{\boldsymbol{\beta}}_1) &= E\left(\left(\mathbf{X}_1^T \mathbf{X}_1\right)^{-1} \mathbf{X}_1^T \mathbf{Y}\right) \\ &= \left(\mathbf{X}_1^T \mathbf{X}_1\right)^{-1} \mathbf{X}_1^T E[\mathbf{Y}] \\ &= \left(\mathbf{X}_1^T \mathbf{X}_1\right)^{-1} \mathbf{X}_1^T (\mathbf{X}_1 \boldsymbol{\beta}_1 + \mathbf{X}_2 \boldsymbol{\beta}_2) \\ &= \boldsymbol{\beta}_1 + \left(\mathbf{X}_1^T \mathbf{X}_1\right)^{-1} \mathbf{X}_1^T \mathbf{X}_2 \boldsymbol{\beta}_2 \end{aligned}$$

Alias optimal design

- Bias

$$E(\hat{\boldsymbol{\beta}}_1) - \boldsymbol{\beta}_1 = (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{X}_1^T \mathbf{X}_2 \boldsymbol{\beta}_2$$

- Estimates are too small or too big
- Alias matrix

$$\mathbf{A} = (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{X}_1^T \mathbf{X}_2$$

- Ideally, the alias matrix is a null matrix (i.e., a matrix full of zeros)
- Alias optimal designs try to achieve this, but ignore information on interactions

Bayesian optimal screening designs

- Primary term model

$$\mathbf{Y} = \mathbf{X}_1 \boldsymbol{\beta}_1 + \varepsilon$$

- Primary and potential term model

$$\mathbf{Y} = \mathbf{X}_1 \boldsymbol{\beta}_1 + \mathbf{X}_2 \boldsymbol{\beta}_2 + \varepsilon$$

- Ordinary optimal design for all terms

$$\begin{bmatrix} \hat{\boldsymbol{\beta}}_1 \\ \hat{\boldsymbol{\beta}}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{X}_1^T \mathbf{X}_1 & \mathbf{X}_1^T \mathbf{X}_2 \\ \mathbf{X}_2^T \mathbf{X}_1 & \mathbf{X}_2^T \mathbf{X}_2 \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{X}_1^T \mathbf{y} \\ \mathbf{X}_2^T \mathbf{y} \end{bmatrix} \quad \text{maximize } \det \begin{bmatrix} \mathbf{X}_1^T \mathbf{X}_1 & \mathbf{X}_1^T \mathbf{X}_2 \\ \mathbf{X}_2^T \mathbf{X}_1 & \mathbf{X}_2^T \mathbf{X}_2 \end{bmatrix}$$

- Bayesian optimal design prioritizing the primary terms

$$\begin{bmatrix} \hat{\boldsymbol{\beta}}_1 \\ \hat{\boldsymbol{\beta}}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{X}_1^T \mathbf{X}_1 & \mathbf{X}_1^T \mathbf{X}_2 \\ \mathbf{X}_2^T \mathbf{X}_1 & \mathbf{X}_2^T \mathbf{X}_2 + \frac{\mathbf{I}}{\tau^2} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{X}_1^T \mathbf{y} \\ \mathbf{X}_2^T \mathbf{y} \end{bmatrix} \quad \text{maximize } \det \begin{bmatrix} \mathbf{X}_1^T \mathbf{X}_1 & \mathbf{X}_1^T \mathbf{X}_2 \\ \mathbf{X}_2^T \mathbf{X}_1 & \mathbf{X}_2^T \mathbf{X}_2 + \frac{\mathbf{I}}{\tau^2} \end{bmatrix}$$

Bayesian optimal screening designs

- Prior distribution for potential terms

$$\beta_2 \sim \text{NORMAL}(\mathbf{0}, \tau^2 \sigma^2 \mathbf{I})$$

this means: “I think these parameters
are not in the model”
or “I think their effects are zero”

this means: “I’m not totally sure of that”

- The larger τ^2 , the more attention you want to pay to the potential terms
- Number of tests can be lower than the total number of terms (primary + potential)

Bayesian optimal screening designs

- Can be used for I-optimal design too
- Drawbacks
 - Generally not orthogonal
 - Optimal design algorithms produce a different design for a given problem each time you run them
- Definitive screening designs (DSDs) and OMARS designs overcome this issue
 - DSDs are Bayesian D-optimal designs (or close) + alias optimal
 - OMARS designs are Bayesian I-optimal designs (or close)

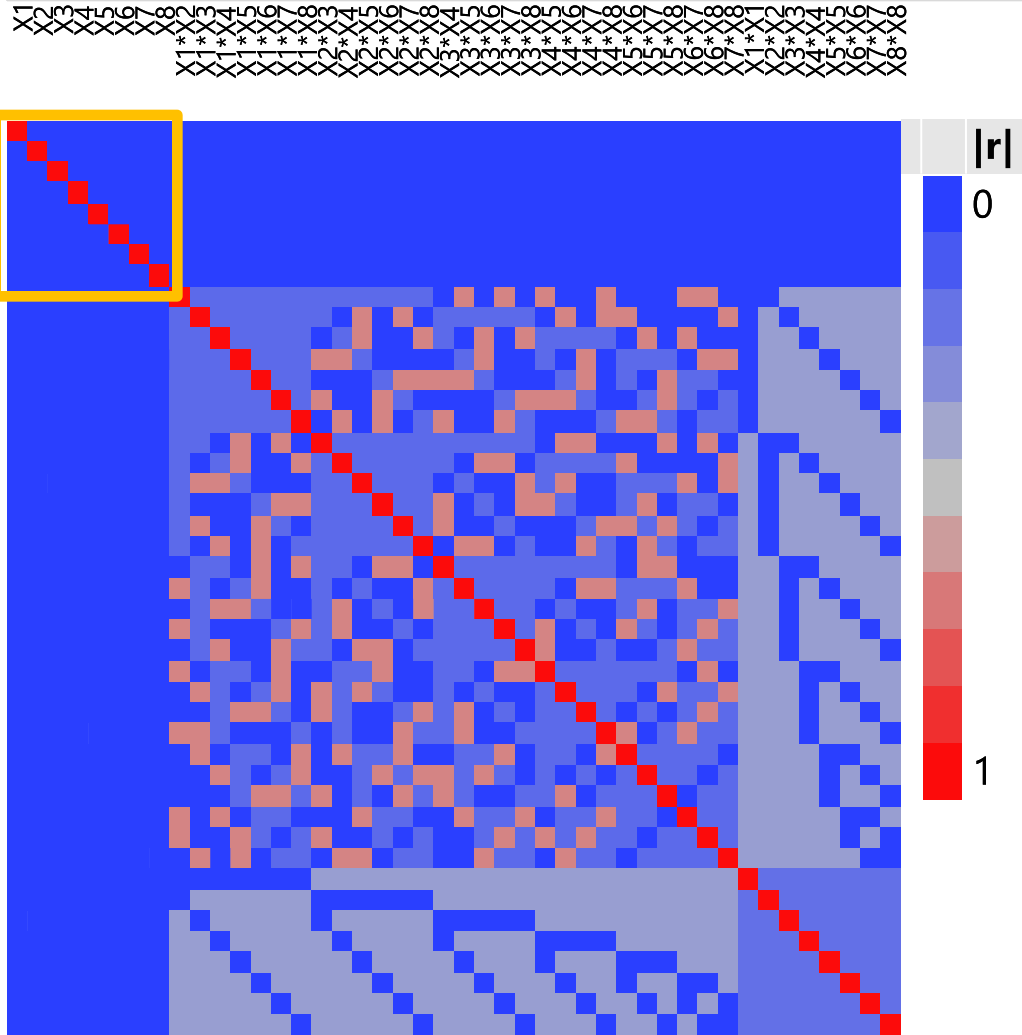
Definitive Screening Designs

Definitive screening design (DSD)

- Jones and Nachtsheim (2011)
- Originally meant for quantitative experimental factors
- Involves three levels for each factor
- Requires few experimental runs when used for a completely randomized experiment
- Initially
 - Algorithmically constructed
 - Number of runs was twice the number of factors, plus 1
- Now
 - combinatorial construction based on conference matrices
 - number of runs is twice the number of factors, plus 1 or 3

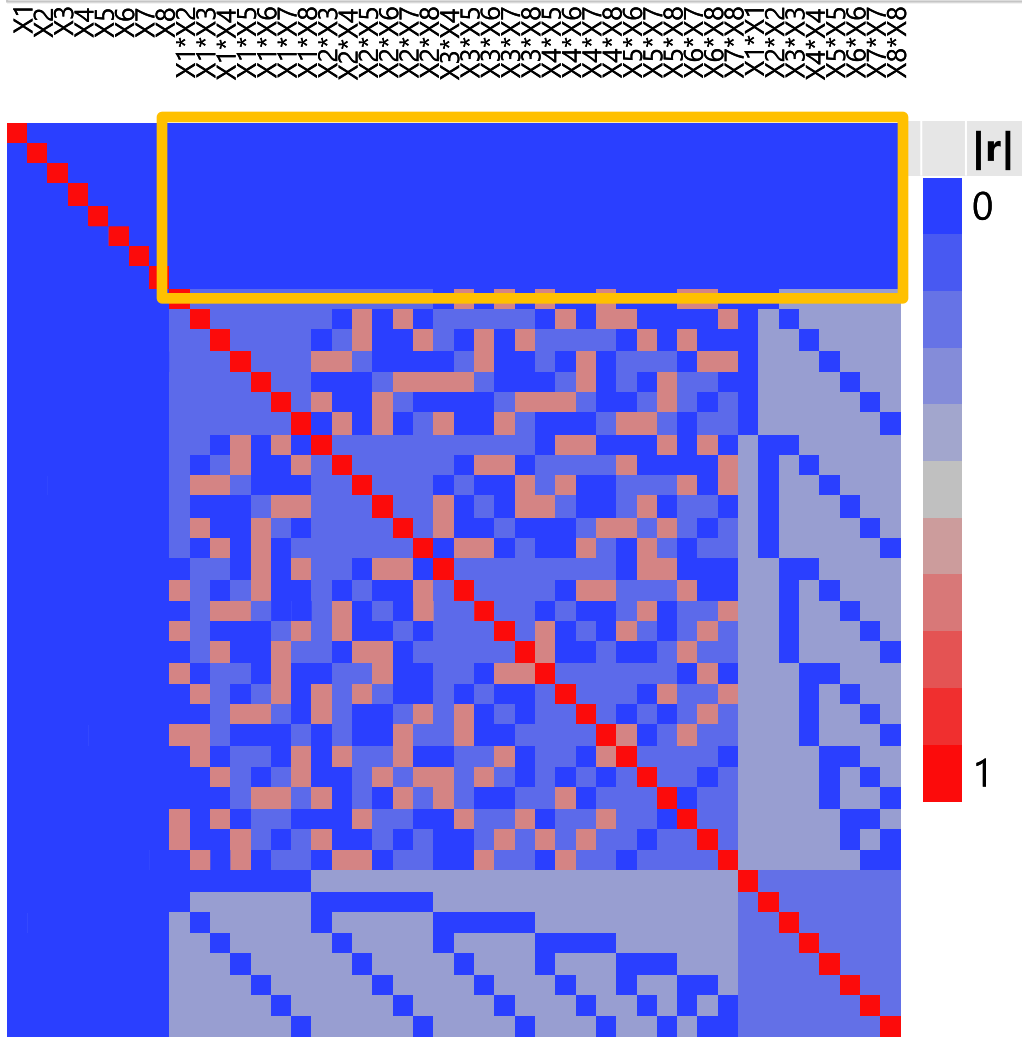
8-factor definitive screening design

Color Map on Correlations



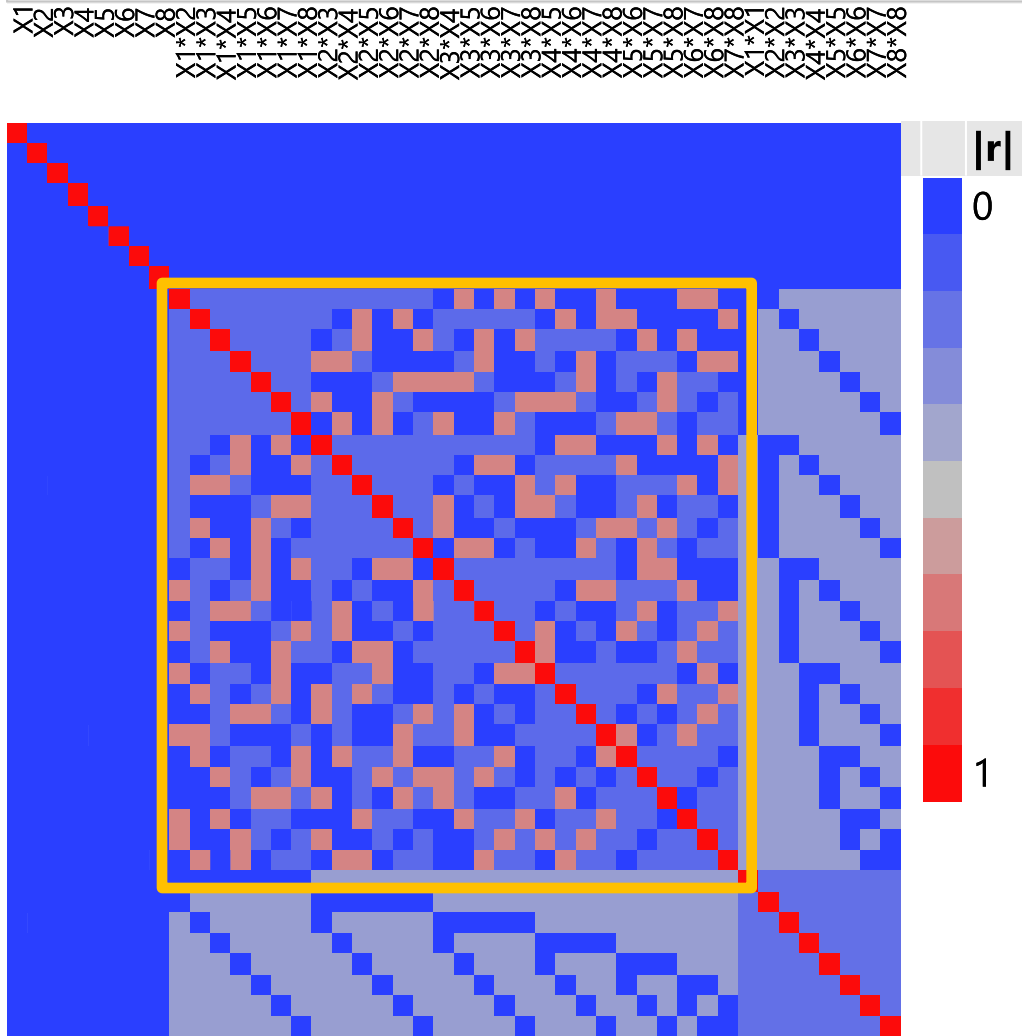
8-factor definitive screening design

Color Map on Correlations



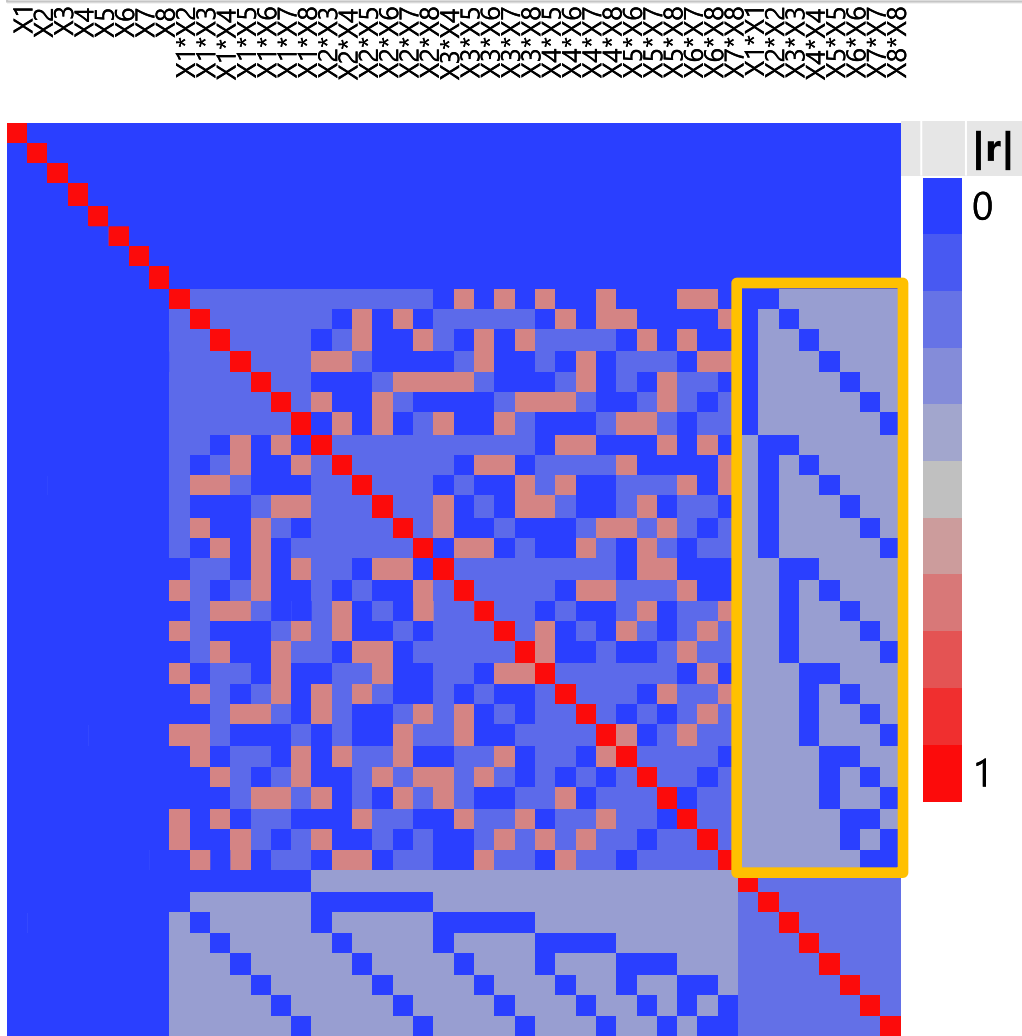
8-factor definitive screening design

Color Map on Correlations



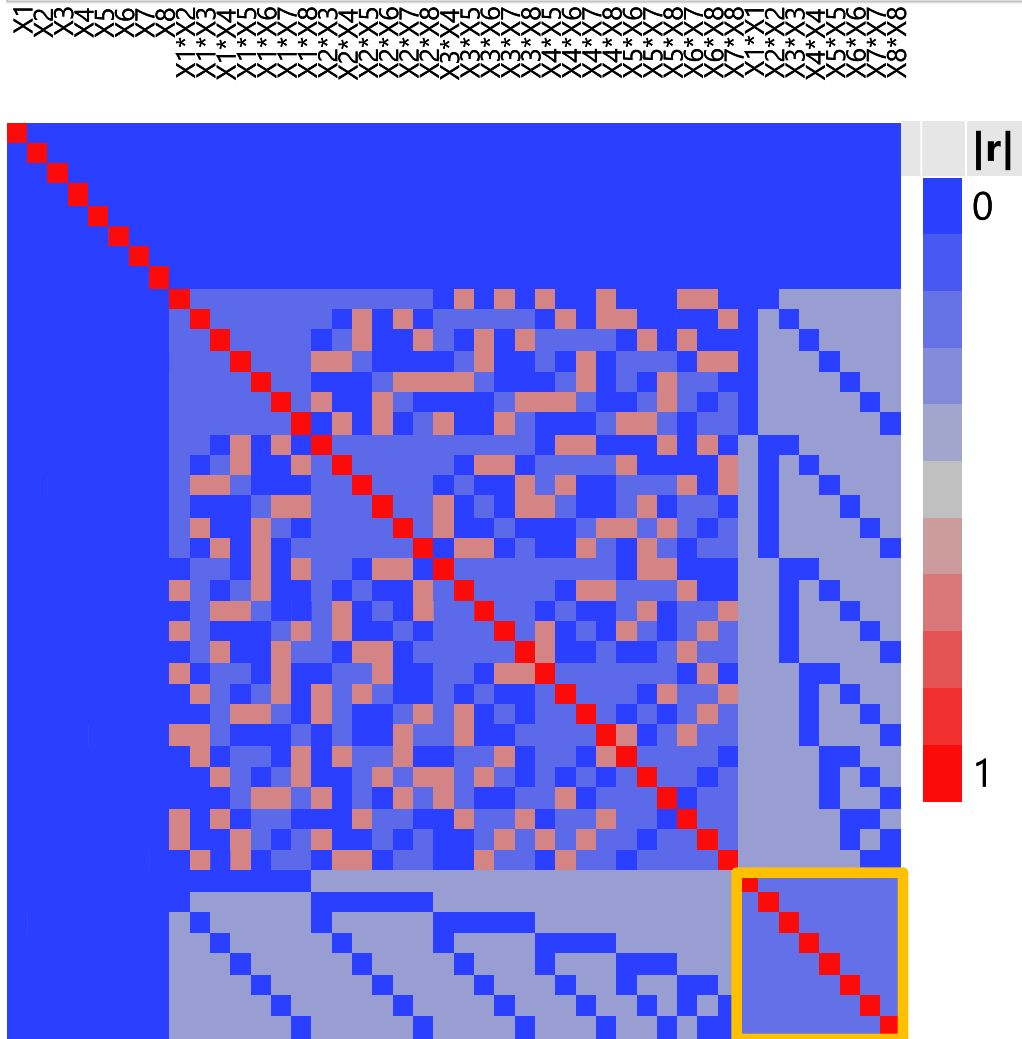
8-factor definitive screening design

Color Map on Correlations



8-factor definitive screening design

Color Map on Correlations



Features

- Every factor is studied at 3 levels
- The design involves as many fold-over pairs as there are factors in the design
- The standard definitive screening design has one center run
- Every factor is studied at its middle level 3 times
- Construction of definitive screening design is nowadays based on a special matrix, called conference matrix

Properties

- Allow precise estimation of main effects
- Allow estimation of all quadratic effects (unlike two-level designs with center runs)
- Allow detection of curvature in the relationship between a quantitative factor and the response
- Main effect estimates are independent from estimates of quadratic effects
- Main effect estimates are not biased if there are active two-factor interactions

Properties

- Two-factor interaction effects are not completely aliased with each other
- If only a few factors matter, a second-order response surface model (incl. quadratic effects and 2-factor interactions) can be estimated precisely
- This basically means DSDs are a screening design and a response surface design in one, if there is factor sparsity

Properties

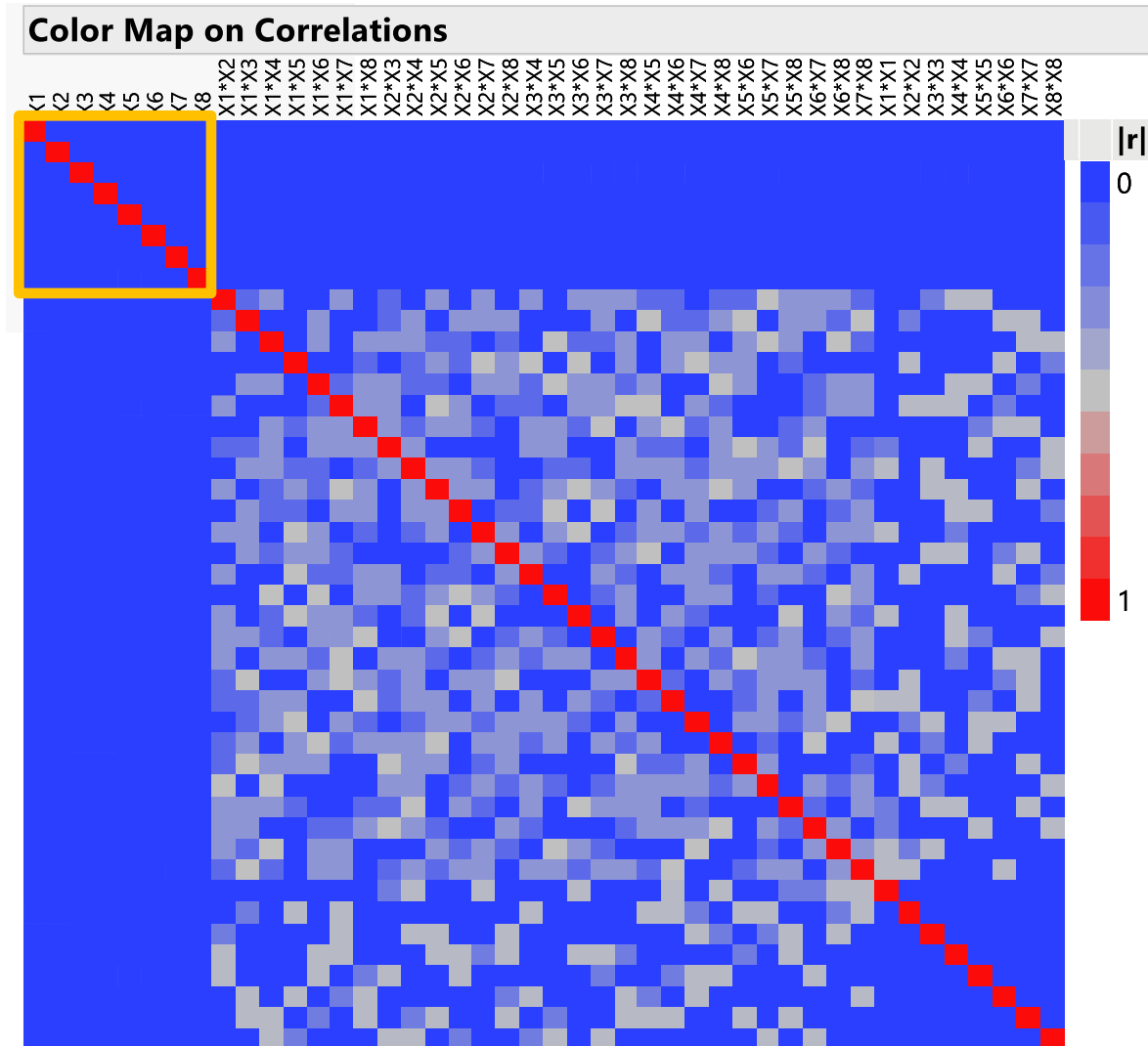
- Can be adapted in case there are two-level categorical factors too (but the DSDs are then no longer orthogonal)
- Can be adapted if the experiment has to be run in blocks (but suboptimal blocking patterns)
- Drawbacks:
 - Perhaps a bit too small
 - Too much aliasing among interactions
 - Too low power for quadratic effects

OMARS Designs

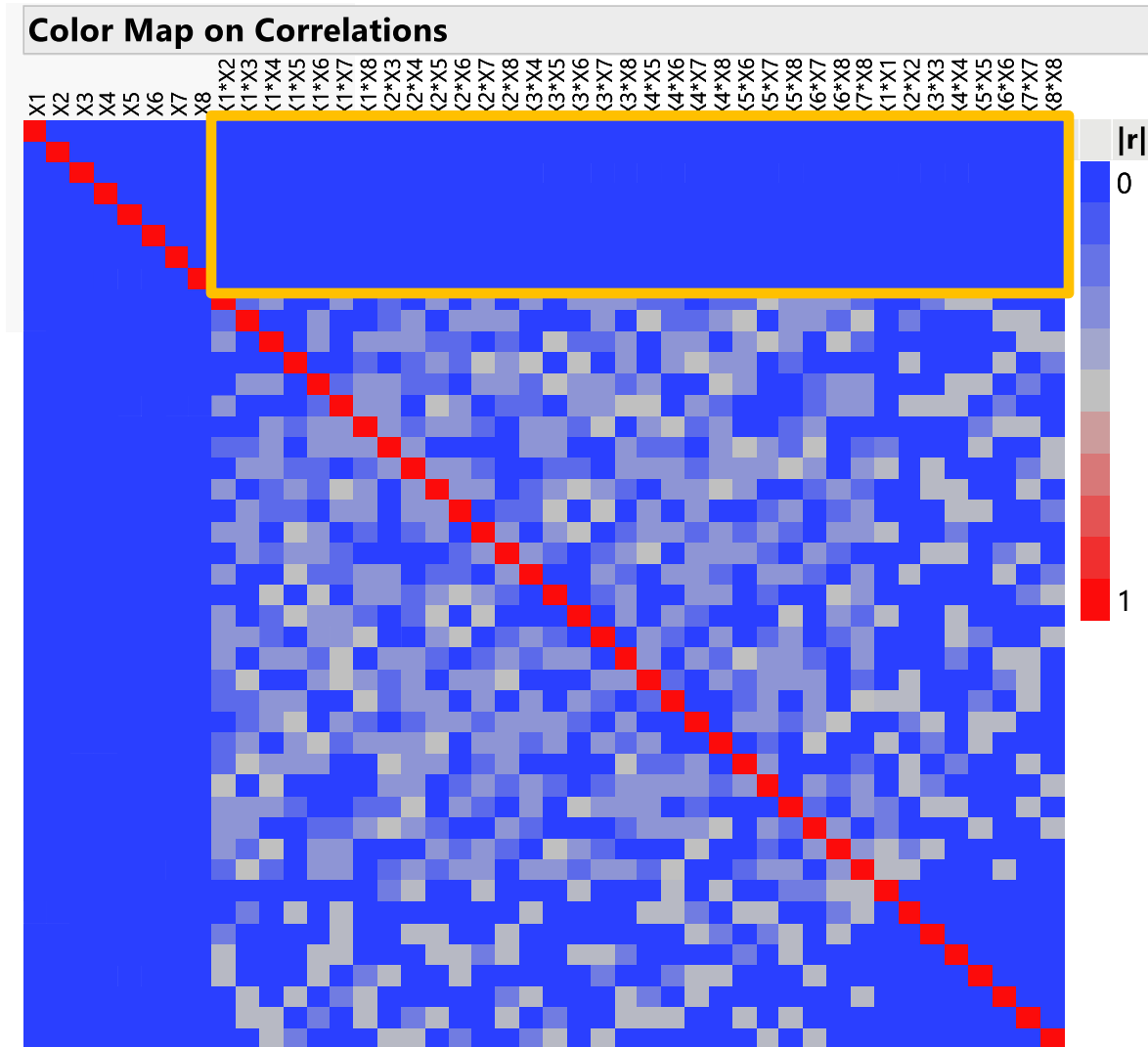
Orthogonal minimally aliased response surface designs

- Original OMARS designs are **three-level designs** for quantitative factors
- Therefore, they are called **response surface designs**
- In OMARS designs, main effects are orthogonal to each other
- Therefore, they are called **orthogonal**
- In OMARS designs, main effects are orthogonal to two-factor interactions and to quadratic effects
- Therefore, they are called **minimally aliased**

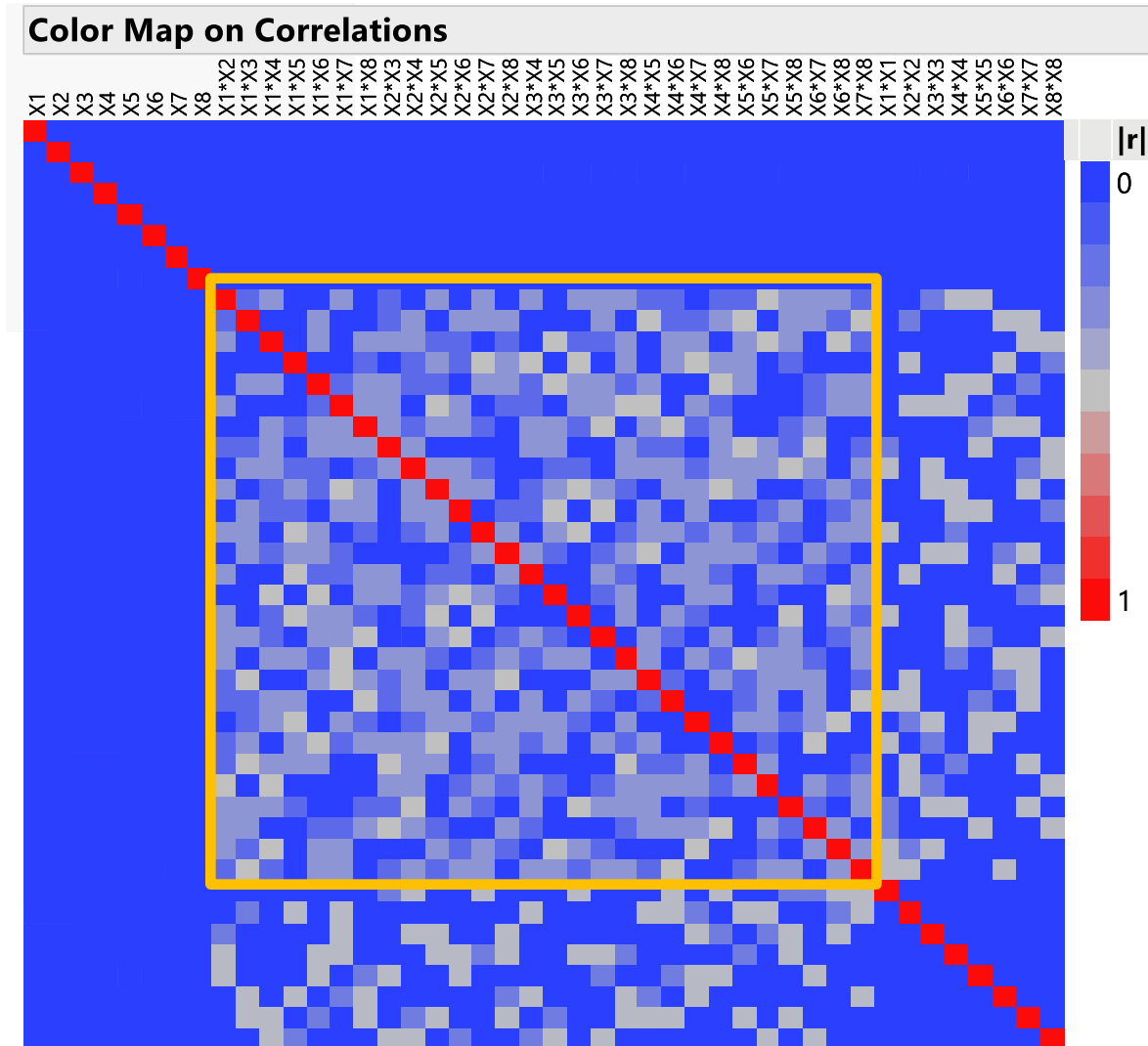
8-factor 27-run OMARS design



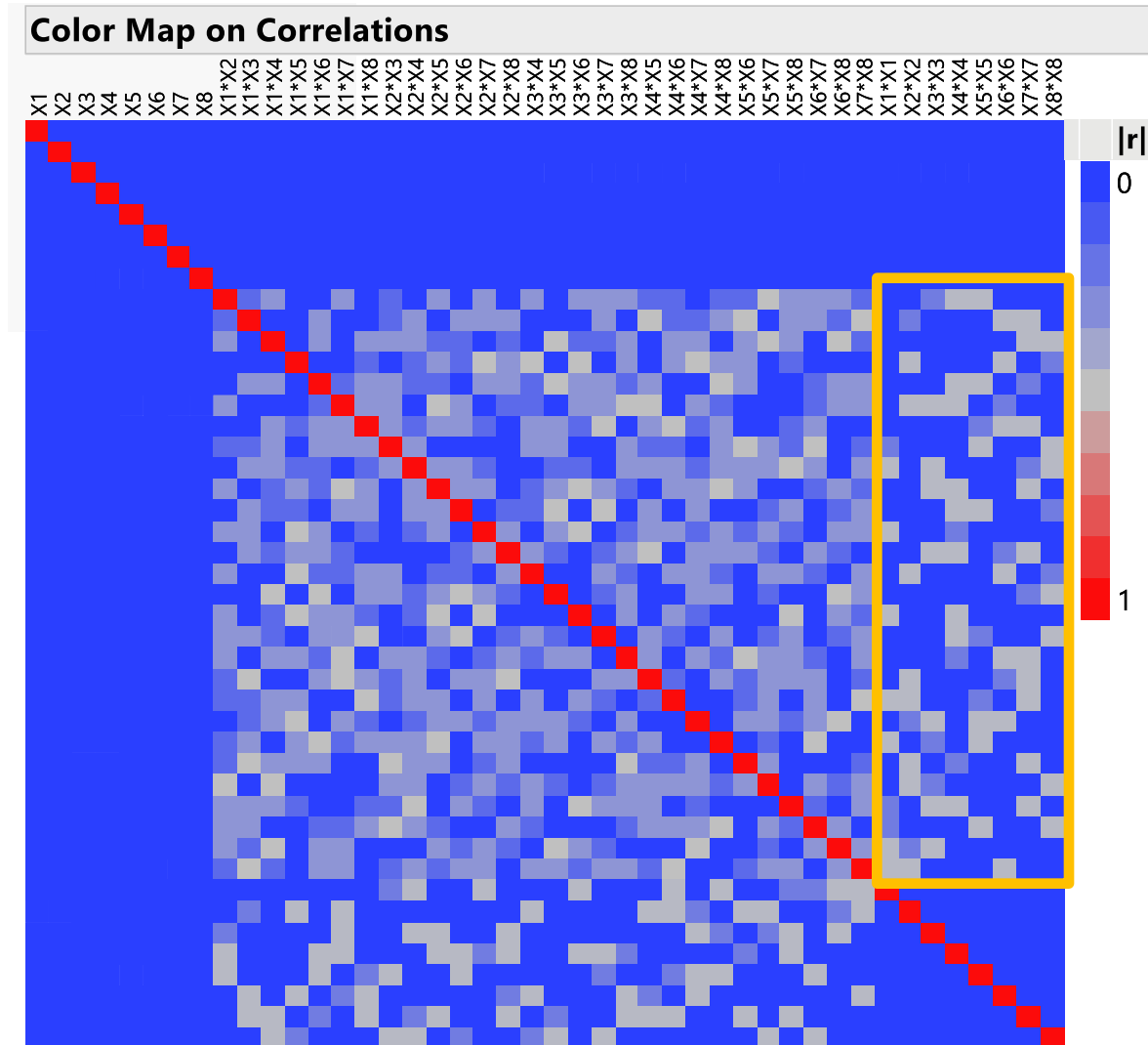
8-factor 27-run OMARS design



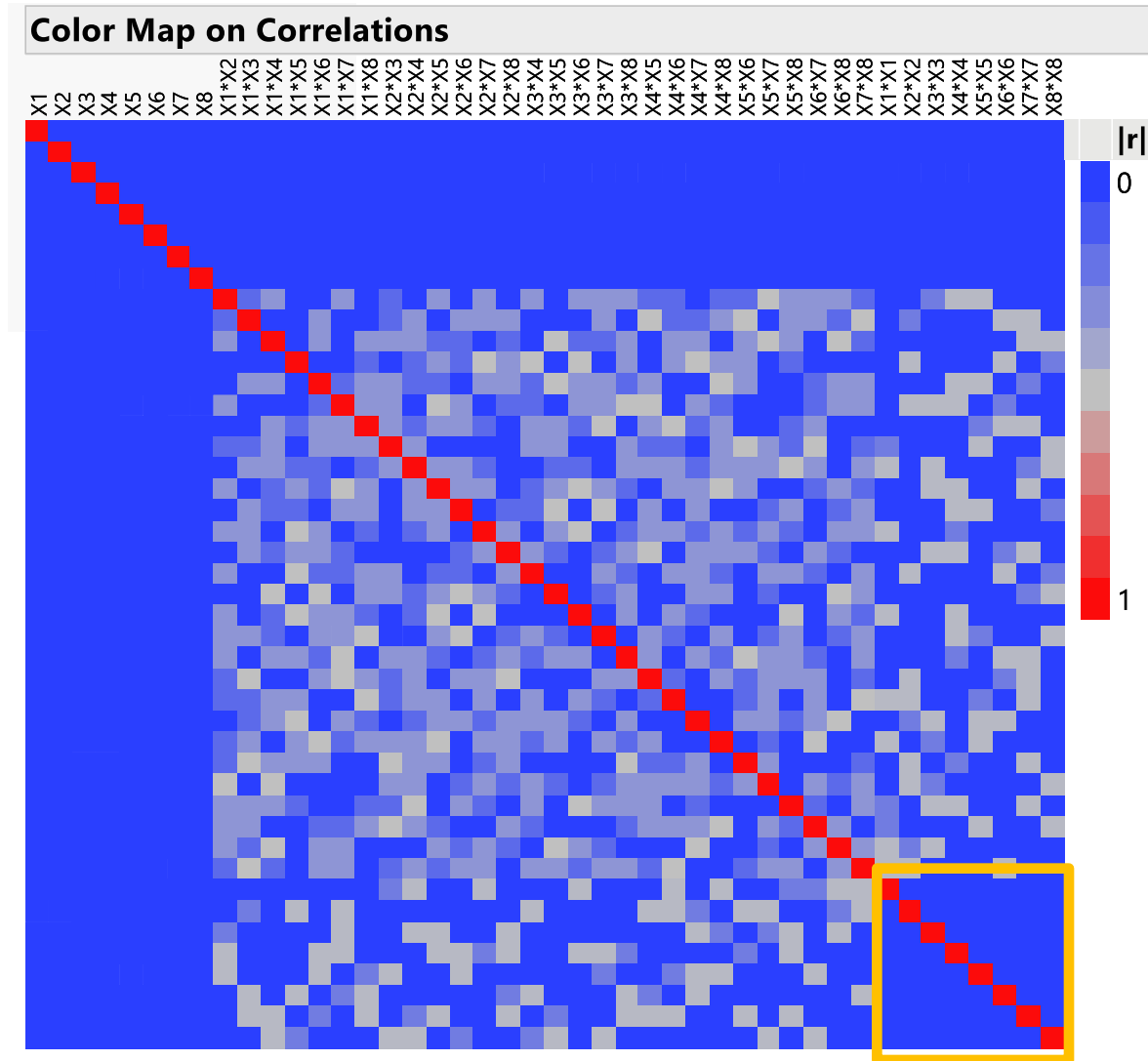
8-factor 27-run OMARS design



8-factor 27-run OMARS design



8-factor 27-run OMARS design



How many OMARS designs exist?

- Special cases
 - Box-Behnken designs
 - Face-centered central composite designs
 - Three-level full factorial designs
 - Definitive screening designs
- Complete enumeration
 - 1000s for up to 5 factors
 - 100,000s and 1,000,000s for 6 or more factors
 - Most are fold-over designs
- Núñez Ares J. & Goos P. (2020) Enumeration and Multi-criteria Selection of Orthogonal Minimally Aliased Response Surface Designs, Technometrics, 62:1, 21-36

- 22 runs
- Foldover design
- Balanced: as many +1s as –1s in every column
- Six 0s in every column
- Center runs can be added

X1	X2	X3	X4	X5
-1	-1	-1	-1	-1
-1	-1	1	0	-1
-1	-1	1	1	1
-1	0	-1	0	0
-1	0	0	-1	1
-1	1	-1	1	1
-1	1	0	-1	0
-1	1	1	1	-1
0	-1	-1	1	0
0	-1	0	0	1
0	0	-1	1	-1
0	0	1	-1	1
0	1	0	0	-1
0	1	1	-1	0
1	-1	-1	-1	1
1	-1	0	1	0
1	-1	1	-1	-1
1	0	0	1	-1
1	0	1	0	0
1	1	-1	-1	-1
1	1	-1	0	1
1	1	1	1	1

New developments

- Our original designs are **uniform precision OMARS designs**, because
 - Main effects can be estimated with same precision
 - Quadratic effects can be estimated with same precision
- Many **non-uniform precision OMARS designs** exist
- Orthogonally blocked OMARS designs
- We also generated **strong OMARS designs**
 - Two-factor interactions orthogonal to other two-factor interactions
 - Two-factor interactions orthogonal to quadratic effects

OMARS designs with 2-level factors

- Many experiments involve categorical factors too
- Definitive screening designs have been extended to incorporate two-level categorical factors
 - Jones & Nachtsheim (2013) presented DSD-based column augmentation and orthogonal column augmentation (former designs are in JMP)
 - Nachtsheim, Shen & Lin (2017) presented compromise designs
- Nguyen, Pham & Mai (2020) presented mixed-level fold-over designs
- None of these designs possesses the orthogonality and minimal aliasing properties

Mixed-level OMARS design 1

1	0	-	+	+	+	+	+	-	13	0	-	+	-	-	-	+
2	0	+	-	-	-	-	-	+	14	0	+	-	+	+	+	-
3	-	0	+	+	+	-	-	-	15	-	0	-	+	-	+	+
4	+	0	-	-	-	+	+	+	16	+	0	+	-	+	-	-
5	-	-	0	-	+	-	+	+	17	-	+	0	-	-	+	-
6	+	+	0	+	-	+	-	-	18	+	-	0	+	+	-	+
7	-	-	-	0	-	-	+	-	19	-	+	+	0	-	-	+
8	+	+	+	0	+	+	-	+	20	+	-	-	0	+	+	-
9	-	+	-	+	+	-	-	+	21	-	-	-	-	+	+	-
10	+	-	+	-	-	+	+	-	22	+	+	+	+	-	-	+
11	-	+	+	-	+	+	+	+	23	-	-	+	+	-	+	+
12	+	-	-	+	-	-	-	-	24	+	+	-	-	+	-	-

Mixed-level OMARS design 2

1	0	-	+	0	0	-	-	-	-	-	-	-	-	-	-	-	25	0	0	0	0	-	+	-	-	-	+	+	+	-	+	+	+	-	+	+	
2	0	+	0	0	0	-	-	-	-	-	-	-	+	+	+	+	26	+	0	0	-	0	+	-	-	-	+	+	+	+	-	-	+	-	-	-	
3	0	0	0	+	0	-	-	-	-	+	+	+	-	-	-	+	27	+	-	-	+	0	+	-	-	+	-	+	+	-	+	+	+	+	+	-	
4	0	0	-	+	+	-	-	-	+	+	-	-	+	-	+	+	28	-	-	0	-	+	+	-	-	+	-	+	+	-	+	-	-	+	+	+	
5	0	0	-	-	-	-	-	+	-	+	-	+	-	+	-	-	+	29	0	0	+	0	0	+	-	-	+	+	-	-	-	+	+	+	-	-	+
6	0	+	0	-	+	-	-	+	+	-	+	+	+	+	+	-	+	30	-	+	+	+	-	+	-	-	+	+	+	-	+	-	-	-	+	+	-
7	-	-	+	0	-	-	+	-	-	+	+	+	+	+	+	+	+	31	-	+	-	-	0	+	-	+	-	-	-	-	+	+	-	+	-	+	-
8	+	0	0	0	0	-	-	+	-	+	+	-	+	+	+	-	+	32	+	+	0	0	0	+	-	+	-	-	+	+	-	-	-	-	+	-	+
9	0	+	+	0	+	-	+	+	-	+	+	-	-	+	+	-	+	33	0	-	0	-	0	+	-	+	-	+	-	-	-	-	+	-	+	+	-
10	-	0	-	0	-	-	+	+	+	-	+	-	-	-	+	+	-	34	-	0	0	+	+	+	-	+	-	+	-	+	+	-	+	-	-	-	+
11	+	0	0	0	-	-	+	+	+	-	+	-	+	-	-	-	+	35	+	0	+	0	0	+	-	+	+	-	-	-	+	+	-	+	+	-	+
12	0	-	0	0	+	-	+	+	+	+	-	+	-	-	+	-	+	36	0	0	-	+	0	+	-	+	+	-	+	-	-	+	+	-	-	-	-
13	0	0	+	-	0	+	-	-	+	-	+	+	-	-	+	+	+	37	0	+	0	0	-	+	+	-	-	-	-	+	-	+	+	-	+	-	-
14	-	0	-	0	0	+	-	-	+	+	+	-	-	+	-	-	+	38	-	0	0	0	+	+	+	-	-	-	+	-	+	-	+	+	+	-	+
15	+	0	0	-	-	+	-	+	-	+	-	-	+	-	+	+	+	39	+	0	+	0	+	+	+	-	-	-	+	-	+	+	+	-	-	+	-
16	0	+	0	+	0	+	-	+	-	+	+	+	+	+	-	+	-	40	0	-	-	0	-	+	+	-	-	+	-	-	+	+	-	-	+	-	+
17	-	-	0	0	0	+	-	+	+	-	-	+	+	+	+	-	+	41	-	0	0	0	0	+	+	+	-	+	-	-	+	-	-	-	+	-	-
18	+	-	+	+	0	+	-	+	+	+	+	-	-	+	-	+	-	42	+	+	-	0	+	+	+	-	+	+	-	-	-	-	-	-	-	+	+
19	+	-	-	-	+	+	+	-	-	-	+	-	+	+	+	-	-	43	0	-	0	+	-	+	+	+	-	-	+	-	-	-	-	+	-	+	+
20	0	0	-	0	0	+	+	+	-	+	+	+	-	-	-	+	+	44	0	0	+	+	+	+	+	+	-	+	-	+	-	+	-	+	+	+	-
21	+	+	0	+	-	+	+	-	+	-	-	+	-	+	+	-	-	45	0	0	+	-	-	+	+	+	+	-	-	+	+	-	+	-	-	+	+
22	-	+	+	-	0	+	+	-	+	+	-	-	+	-	-	-	-	46	0	0	0	-	0	+	+	+	+	+	+	-	-	-	+	+	+	-	-
23	-	0	0	+	0	+	+	+	-	-	-	-	+	+	-	+	+	47	0	-	0	0	0	+	+	+	+	+	+	+	+	+	+	-	-	-	-
24	0	0	0	0	+	+	+	+	-	-	-	+	-	-	-	+	-	48	0	+	-	0	0	+	+	+	+	+	+	+	+	+	+	+	+	+	+

A non-uniform precision OMARS design

X1	X2	X3	X4	X1	X2	X3	X4
1	-1	-1	1	1	0	0	0
0	1	0	1	0	-1	-1	0
0	-1	1	0	0	-1	1	-1
-1	0	1	1	-1	0	1	-1
-1	1	-1	-1	1	1	-1	0
1	-1	1	1	0	1	-1	1
1	0	-1	-1	-1	1	1	0
-1	0	-1	1	1	0	0	0
-1	-1	-1	-1	-1	-1	0	1
-1	-1	0	0	1	1	1	1
1	1	1	-1	1	-1	0	-1
-1	1	0	0	0	1	0	-1



GSK's first OMARS design application

The case study

- Goal: find relevant factors and optimal settings to increase biomass
- Factors:
 - Holding Time
 - Inoculum Dilution
 - Iron
 - pH
 - Temperature
 - Speed Pump
- The Ambr250 system automatically manages 24 bioreactors (250ml) at the same time

NOTE:

1 intercept

6 main effects

6 quadratic effects

$6 \times 5 / 2 = 15$ interactions

So 28 parameters

Optimal design approach
infeasible

A powerful combo: OMARS designs and Ambr250 system



The selected OMARS design

X1	X2	X3	X4	X5	X6
-1	1	1	-1	-1	-1
			-1	1	0
			1	-1	1
			1	0	-1
			-1	1	1
			1	0	0
			1	0	0
					1
					-1
					1
					0
					0
					0
					0
	0	0			0
0	0				0
0	1				-1
1	-1				1
1	-1				-1
1	-1		1		-1
			1		1
			1		1
			-1		1
			-1		1
			-1		1
			1		0
			1	1	1

6 Process Parameters

Center Points X 2

No Blocks

24 runs

- 24 runs
- 3 duplicated test combinations (one of which is the center point)
- Fold-over design
- Non-uniform precision
 - X1-X4 have 6 zeros
 - X5 has 8 zeros
 - X6 has 10 zeros



WLGore's Stent Adhesive Experiment

Application at WLGore

Step	Factor	Unit	Low	High	No. Levels	Symbol
Polymerization	CONFIDENTIAL				3	X1
Polymerization					3	X2
Coagulation					2	X3
Coagulation					2	X4
Coagulation	Coag Temp	°C	40	65	3	X5
Coagulation	Wash Temp	°C	50	85	3	X6
Coagulation	No. Cycles		6	12	2	X7
Drying	Drying Temperature	°C	130	150	3	X8

OMARS design

X1	X2	X3	X4	X5	X6	X7	X8
-1	1	1	-1	0	0	1	-1
0	1	-1	-1	1	0	-1	-1
-1	-1	-1	-1	1	1	1	0
1	-1	-1	1	0	0	-1	1
1	1	1	1	1	1	1	1
-1	0	-1	1	-1	0	1	1
1	0	1	-1	1	0	-1	-1
1	-1	-1	1	1	-1	1	-1
0	-1	1	-1	0	-1	-1	1
1	0	-1	-1	0	1	-1	1
-1	-1	1	1	1	1	-1	0
-1	-1	-1	-1	-1	-1	-1	-1
1	1	1	1	-1	-1	-1	0
-1	1	-1	1	1	-1	-1	1
0	1	-1	1	0	1	1	-1
0	-1	1	1	-1	0	1	1
-1	1	1	-1	-1	1	-1	1
1	1	-1	-1	-1	-1	1	0
0	0	-1	-1	0	0	1	1
0	0	-1	1	-1	1	-1	-1
0	0	1	-1	1	-1	1	1
0	0	1	1	0	0	-1	-1
1	-1	1	-1	-1	1	1	-1
-1	0	1	1	0	-1	1	-1

Blocked Experiments with Definitive Screening Designs & OMARS Designs

Example 1

Definitive Screening Design - JMP Pro

File Edit Tables Rows Cols DOE Analyze Graph Tools View Window Help

Definitive Screening Design

Design Definitive Screening Design

- Fit Definitive Screening
- Evaluate Design
- DOE Dialog

Columns (6/0)

Block *

X1 *

X2 *

X3 *

Rows

All rows 15

Selected 0

Excluded 0

Hidden 0

Labeled 0

	Block	X1	X2	X3	X4	Y
1	1	-1	-1	1	0	
2	1	0	1	1	1	
3	1	0	-1	-1	-1	
4	1	1	1	-1	0	
5	1	0	0	0	0	
6	2	1	1	1	-1	
7	2	-1	0	1	-1	
8	2	1	0	-1	1	
9	2	-1	-1	-1	1	
10	2	0	0	0	0	
11	3	1	-1	0	-1	
12	3	1	-1	1	1	
13	3	-1	1	-1	-1	
14	3	-1	1	0	1	
15	3	0	0	0	0	

Our blocked DSD

omars design - JMP Pro

File Edit Tables Rows Cols DOE Analyze Graph Tools View Window Help

omars design

Design Definitive Screening Design

- Fit Definitive Screening
- Evaluate Design
- DOE Dialog
- Evaluate Design paper

Columns (6/0)

Block *

- X1 *
- X2 *
- X3 *
- X4 *

Rows

All rows 15

Selected 0

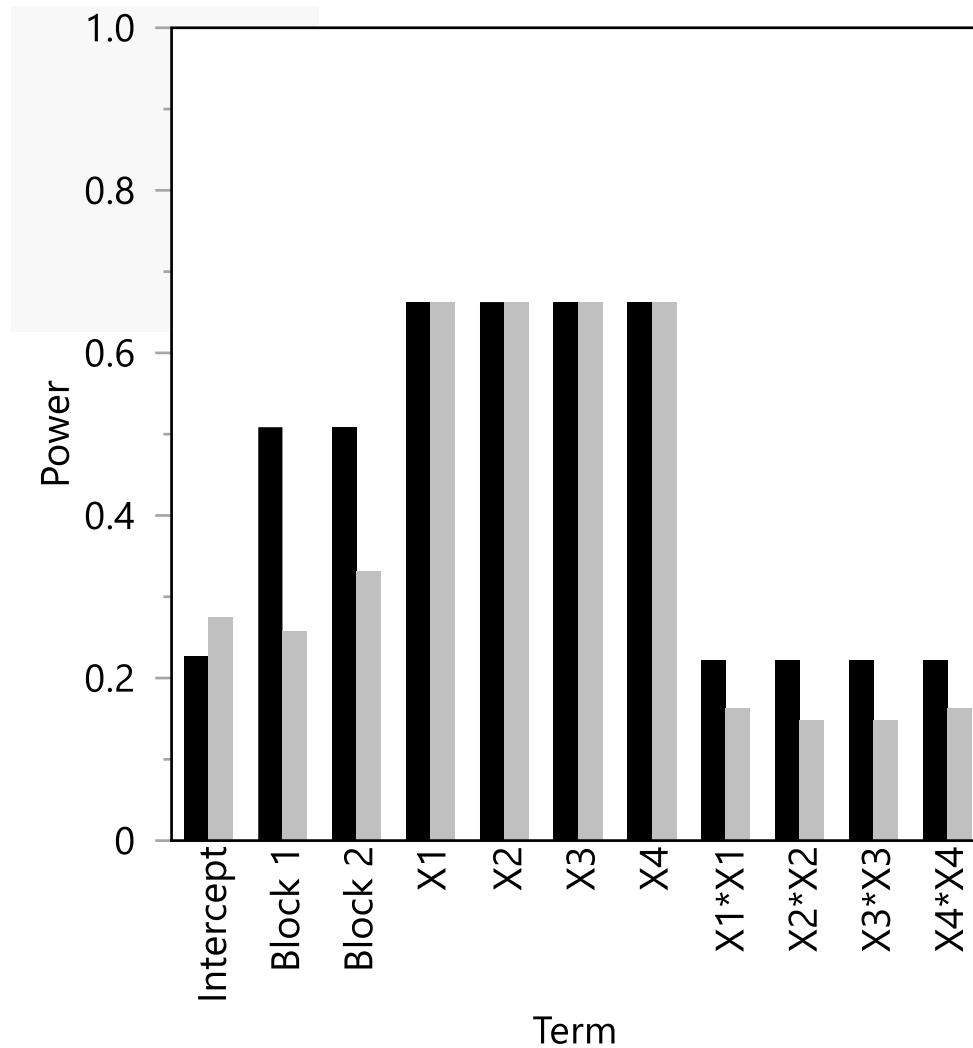
Excluded 0

Hidden 0

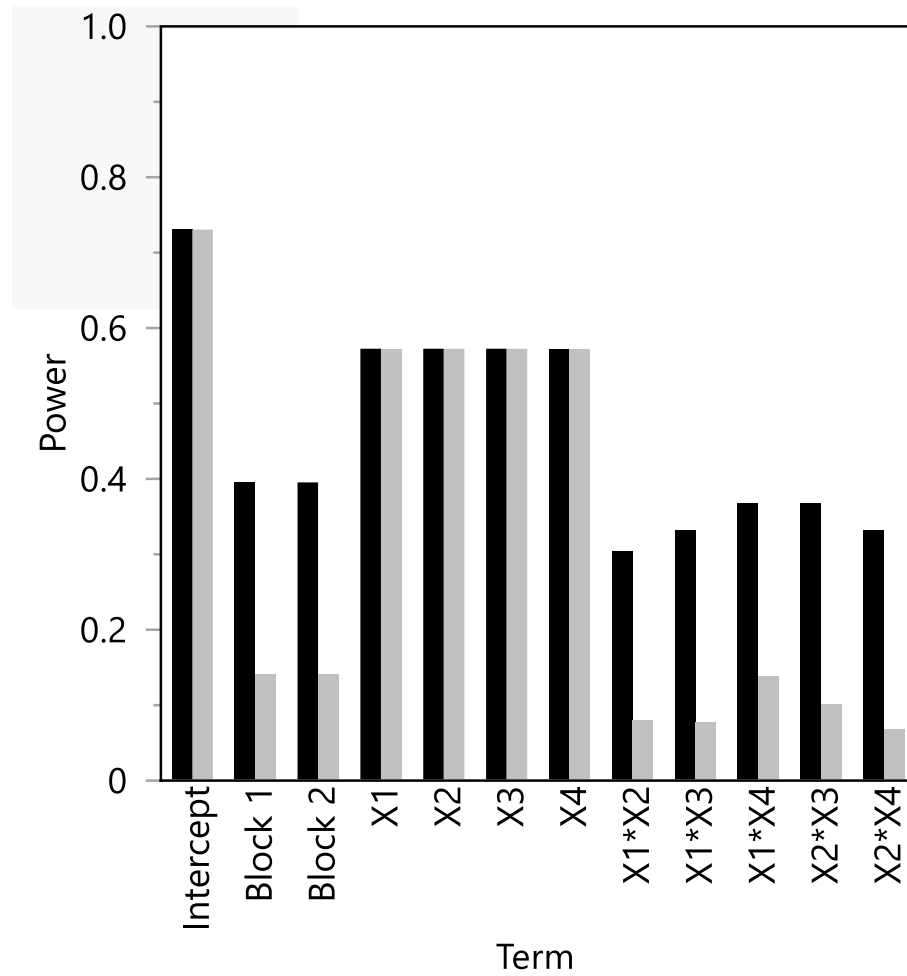
Labeled 0

	Block	X1	X2	X3	X4
1	1	-1	0	1	-1
2	1	-1	1	0	1
3	1	0	-1	-1	1
4	1	1	-1	1	0
5	1	1	1	-1	-1
6	2	-1	-1	1	1
7	2	-1	1	-1	0
8	2	0	1	1	-1
9	2	1	-1	0	-1
10	2	1	0	-1	1
11	3	-1	-1	-1	-1
12	3	0	0	0	0
13	3	0	0	0	0
14	3	1	1	1	1
15	3	0	0	0	0

Power comparison



Power comparison



Example 2

jmp-dsd-5f-25r-5b - JMP Pro

File Edit Tables Rows Cols DOE Analyze Graph Tools View Window Help

jmp-dsd-5f-25r-5b

Design Definitive Screening

- Fit Defini...Screening
- Evaluate Design
- DOE Dialog

Columns (7/0)

Block *

X1 *

X2 *

X3 *

X4 *

X5 *

Y *

Rows

All rows 25

Selected 0

Excluded 0

Hidden 0

Labeled 0

	Block	X1	X2	X3	X4	X5	Y
1	1	0	0	0	0	0	
2	1	0	1	1	1	1	
3	1	-1	1	-1	-1	1	
4	1	0	-1	-1	-1	-1	
5	1	1	-1	1	1	-1	
6	2	-1	-1	1	-1	1	
7	2	0	0	0	0	0	
8	2	-1	0	1	1	1	
9	2	1	1	-1	1	-1	
10	2	1	0	-1	-1	-1	
11	3	1	1	-1	1	1	
12	3	-1	1	0	1	-1	
13	3	-1	-1	1	-1	-1	
14	3	0	0	0	0	0	
15	3	1	-1	0	-1	1	
16	4	-1	-1	-1	1	1	
17	4	-1	1	1	0	-1	
18	4	0	0	0	0	0	
19	4	1	1	1	-1	-1	
20	4	1	-1	-1	0	1	
21	5	0	0	0	0	0	
22	5	-1	-1	-1	1	-1	
23	5	1	1	1	-1	1	
24	5	1	-1	1	1	0	
25	5	-1	1	-1	-1	0	

DSD from JMP

omars-5f-25r-5b - JMP Pro

File Edit Tables Rows Cols DOE Analyze Graph Tools View Window Help

omars-5f-25r-5b

Design Definitive Screening

- Fit Defini...Screening
- Evaluate Design
- DOE Dialog

Columns (7/0)

Block *

X1 *

X2 *

X3 *

X4 *

X5 *

Y *

Rows

All rows 25

Selected 0

Excluded 0

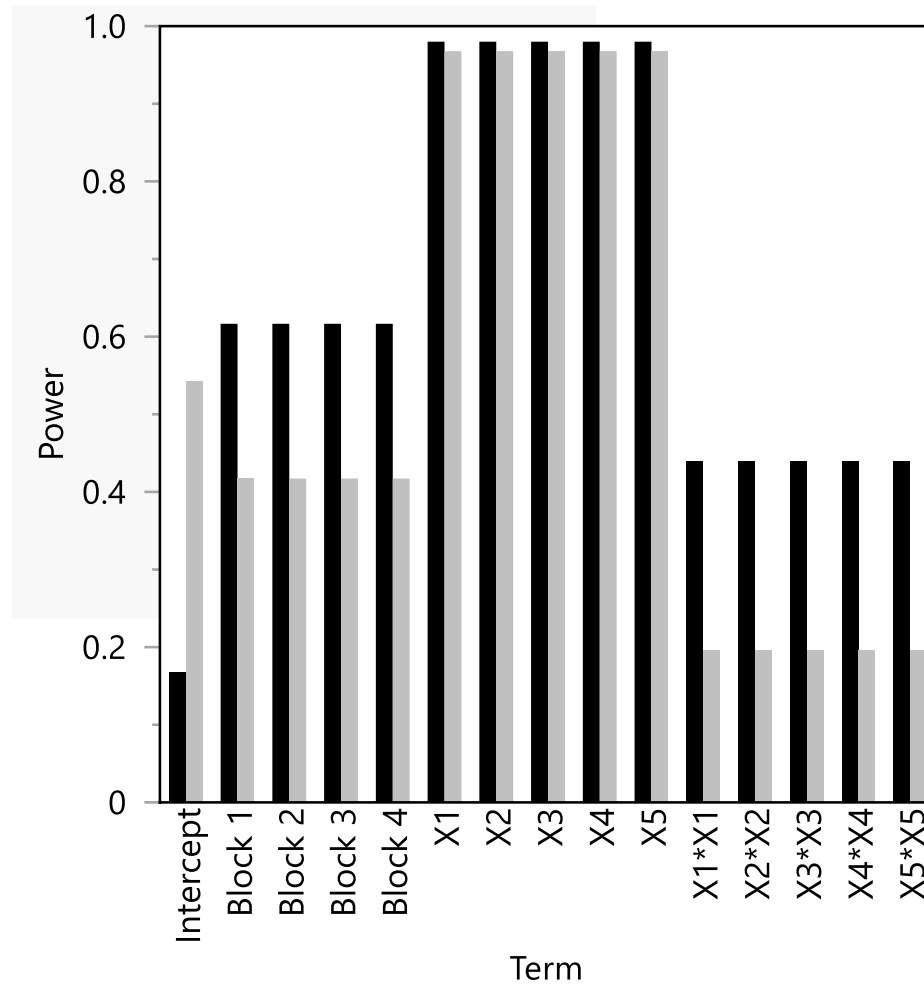
Hidden 0

Labeled 0

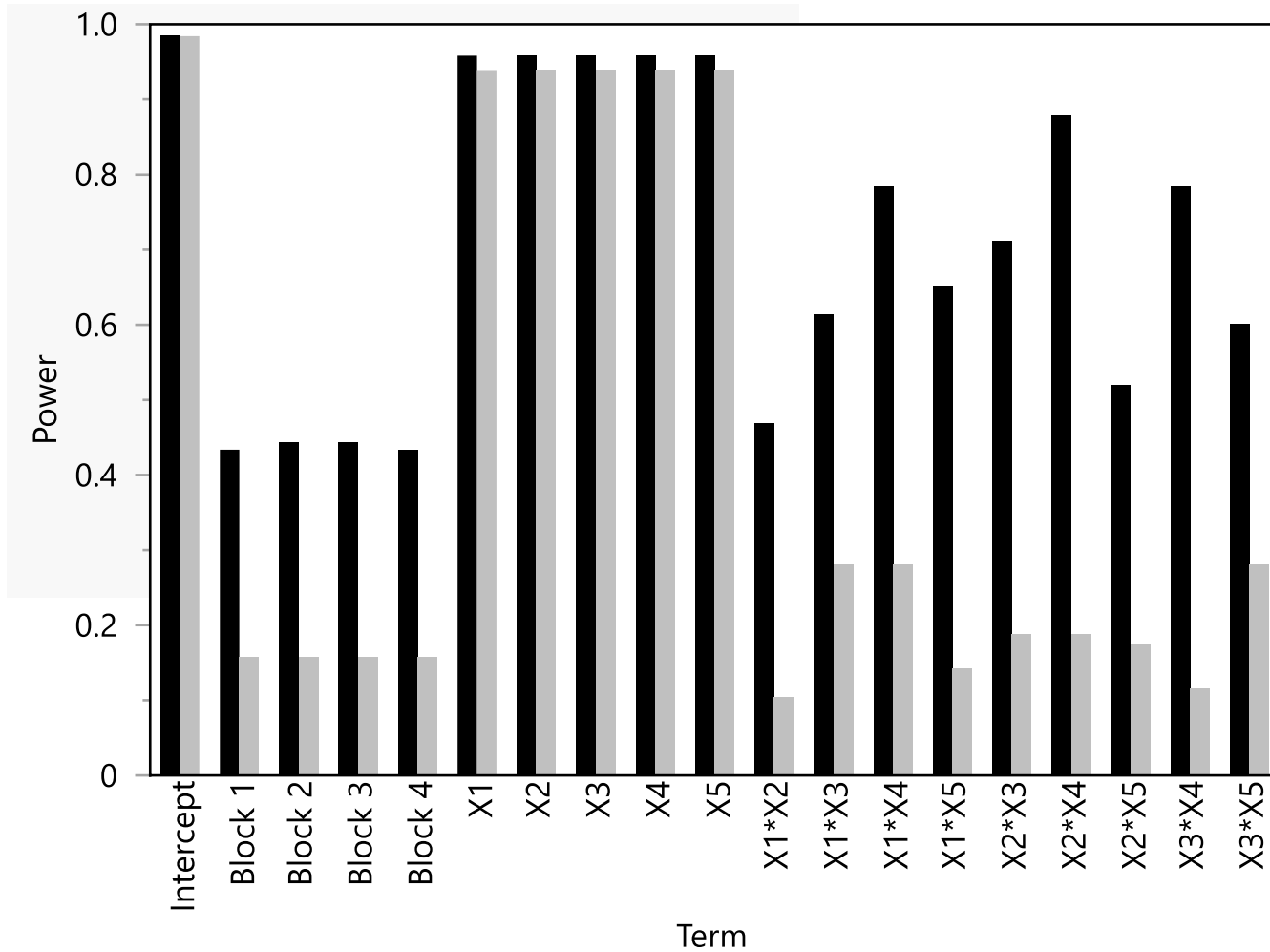
	Block	X1	X2	X3	X4	X5	Y
1	1	1	1	-1	1	0	
2	1	-1	-1	-1	0	-1	
3	1	0	1	1	-1	-1	
4	1	-1	0	1	1	1	
5	1	1	-1	0	-1	1	
6	2	1	-1	-1	0	1	
7	2	-1	-1	0	1	-1	
8	2	1	1	1	1	0	
9	2	-1	0	1	-1	1	
10	2	0	1	-1	-1	-1	
11	3	-1	-1	-1	-1	0	
12	3	1	0	-1	1	-1	
13	3	0	-1	1	1	1	
14	3	-1	1	1	0	-1	
15	3	1	1	0	-1	1	
16	4	1	0	-1	-1	-1	
17	4	-1	1	0	1	-1	
18	4	0	-1	-1	1	1	
19	4	1	1	1	0	1	
20	4	-1	-1	1	-1	0	
21	5	1	-1	1	1	-1	
22	5	1	-1	1	-1	-1	
23	5	0	0	0	0	0	
24	5	-1	1	-1	1	1	
25	5	-1	1	-1	-1	1	

OMARS design

Power comparison



Power comparison



Availability

Minitab



Design of Experiments
Minitab DOE™ by Effex